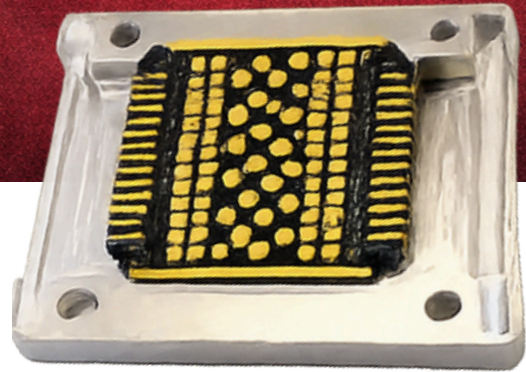


Cryogenic RF Low-Pass Filters

Low-Noise Filtering for Cryogenic Quantum & Measurement Systems



Ultra-Low Noise RF Filtering.
Optimized for Cryogenic Environments.
Precision Signal Conditioning for Quantum Systems

CryoClean cryogenic RF low-pass filters are engineered to suppress high-frequency noise on cryogenic measurement lines while preserving low-frequency signal fidelity. Designed for quantum transport and quantum computing environments, the filters protect sensitive devices from RF interference and electrostatic disturbances at temperatures down to millikelvin regimes. The architecture ensures stable, repeatable filtering performance across thermal cycles and extreme operating conditions.

KEY FEATURES

- **Noise Suppression Performance**
Provides broadband RF and microwave noise attenuation for improved signal purity in sensitive measurements.
- **Cryogenic Stability**
Maintains stable filtering performance down to ~2 K across repeated thermal cycling.
- **High Out-of-Band Attenuation**
Delivers >80 dB attenuation above cutoff frequency to minimize environmental and electromagnetic interference.
- **Low Insertion Loss**
Preserves low-frequency signal amplitude with minimal distortion for accurate quantum transport measurements.
- **Compact Integration**
Compact in-line design compatible with cryostat wiring stacks, probe assemblies, and custom cryogenic setups.

Notes

- Designed for integration within cryogenic wiring environments
- Optimal performance achieved when installed close to the device
- Filtering characteristics depend on selected cutoff configuration
- Custom filter profiles available for specific experimental requirements

APPLICATIONS

- Quantum transport measurements
- Quantum computing device protection (qubits, superconducting circuits)
- Low-temperature nanoelectronics testing
- Cryogenic signal conditioning
- Noise-sensitive electrical measurements



Cryogenic RF Low-Pass Filters

SPECIFICATIONS

ELECTRICAL PERFORMANCE

Filter Type RF low-pass

Frequency Range Application-dependent (customizable cutoff)

Out-of-Band Attenuation > 80 dB

Insertion Loss Low (optimized for signal fidelity)

Signal Type Low-frequency / DC measurement lines

CRYOGENIC PERFORMANCE

Operating Temperature Down to ~2 K

Thermal Stability Stable across repeated thermal cycling

Performance Drift Minimal across temperature range

MECHANICAL & INTEGRATION

Form Factor Compact module

Mounting In-line / wiring stack integration

Integration Probe-compatible / custom setups

Construction Cryogenic-compatible materials

Cryonano Labs | Head Office: Sector V, WTL Building, BP-5, BP Block, Sector V, Bidhannagar, Kolkata- 700091
Bangalore Office: #999/12, 2nd Floor, GMN Complex, 5th 'A' Cross, 1st Block, HRBR Layout, Babusapalya, Banaswadi,
Outer Ring Road, Bangalore – 560043.
www.cryonano.com | Phone: 097481 81485

Cryonano is not responsible for omissions or errors. Specifications subject to change without notice.
© 2026 CryoNano Instruments Pvt. Ltd. All rights reserved.

